

Test 1 Review

1 Prerequisite Practice

Problem 1. Find the equation of the tangent line to the curve $\sqrt{x} + e^{3x}$ at $x = 1$

Problem 2. Solve the system of linear equations: $2x - 7y = 11$ and $x + 3y = 4$

Problem 3. let $f(x) = \sin^{-1}(x)$ what is the $\lim_{x \rightarrow 0} f(x)$, what is the domain of $f(x)$, what is the derivative of $f(x)$?

2 Introduction to Vectors

Problem 4. If $\vec{u} = 5\hat{i} + 3\hat{j} - 2\hat{k}$ what is the magnitude of $\vec{w} = -2\vec{u}$. What is a unit vector in the same direction as w ? Is $\langle -20, -12, 8 \rangle$ parallel to $\vec{u}$ and $\vec{w}$?

Problem 5. Let $\vec{a} = \langle 1, -3, -3 \rangle$ and $\vec{b} = \langle -5, 2, 1 \rangle$. Find the following:

1. $2\vec{a} \cdot \vec{b}$
2. $\vec{a} \times \vec{b}$ and $\vec{b} \times \vec{a}$.
3. $\text{comp}_{\vec{b}}(\vec{a})$ and $\text{proj}_{\vec{b}}(\vec{a})$
4. The angle between $\vec{a}$ and $\vec{b}$
5. The area of the triangle formed by $\vec{a}$ and $\vec{b}$.

Problem 6. Find the direction cosines to the vector $\langle -2, -3, 1 \rangle$

3 Lines in 3 Dimensions

Problem 7. Find the equation of a line which goes through $P(4, 4, 2)$ in the direction $\langle \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \rangle$.

Problem 8. Given the line L written in parametric equation as $x(t) = 2t - 7$, $y(t) = 12 - 5t$ and $z(t) = t + 4$, write L in symmetric form. Where will L cross the plane $y = 0$?

Problem 9. Find the equation of the line through the point $P(-1, -2, 4)$ which is parallel to the line $x(t) = t$, $y(t) = -5t + 1$, $z(t) = 2t + 2$.

Problem 10. Find an equation of any line which is orthogonal to the plane $2(x - 3) + 6(y + 2) - 4(z - 1) = 0$.

Problem 11. Will L_1 and L_2 intersect? If not, are they parallel or skew?

$$L_1 : x = t + 1, y = t + 2, z = 2t + 3$$

and

$$L_2 : x = 3t - 2, y = 5t, z = 3t + 1$$

4 Planes in 3 Dimensions

Problem 12. Find the equation of the plane which goes through $P(10, 20, 30)$ which is normal to the vector $5\hat{i} - \hat{j} - \hat{k}$.

Problem 13. Find the equation of the plane containing the three points $P_1(1, 2, 3)$, $P_2(2, 4, 6)$, and $P_3(-1, -3, -5)$.

Problem 14. Find the distance from the plane $2x - 5y + z = 0$ and the point $P(-2, -7, 3)$.

Problem 15. Find the equation of the plane which contains the point $P(4, -1, -1)$ and the line $\frac{x}{2} = \frac{y-3}{3} = z - 1$

5 Introduction to Vector Functions

Problem 16. What is the domain of the vector function $\langle \frac{1}{t}, \frac{1}{t+1}, \frac{1}{t+2} \rangle$?

Problem 17. What is the domain of the vector function $\langle \frac{1}{e^t-1}, \sqrt[3]{t+5}, \ln(t+5) \rangle$

Problem 18. What is the domain of the vector function $\frac{1}{\sin(t)}\hat{i} + \frac{1}{\cos(t)}\hat{j} + \tan^{-1}(t)\hat{k}$

Problem 19. Sketch the space curve given by the vector function $\vec{r}(t) = \sin^2(t)\hat{i} + \cos^2 t\hat{j} + t\hat{k}$

6 Calculus on Vector functions

Problem 20. Find $\lim_{t \rightarrow 2} \vec{r}(t)$ where,

$$\vec{r}(t) = \left\langle \frac{e^t - e^2}{t^2 - 4}, \cos\left(\frac{\pi t}{2}\right), t + t^2 + t^3 \right\rangle$$

Problem 21. Find $\lim_{t \rightarrow \infty} \vec{r}(t)$ where

$$\vec{r}(t) = \left\langle \frac{\sqrt{t^4 - 2t + 7}}{3t^2 - 9}, e^{-\sqrt{t}}, \frac{\ln t}{t} \right\rangle$$

Problem 22. Find $\vec{r}'(t)$ where $\vec{r}(t) = \sec(t)\hat{i} + \frac{1}{t^5}\hat{j} - 2^t\hat{k}$

Problem 23. Find $\vec{r}(t)$ where $\vec{r}'(t) = \langle 2t, 3t^2, 4t^3 \rangle$ where $\vec{r}(1) = \langle 3, 5, 7 \rangle$

Problem 24. Find $\vec{r}(t)$ where $\vec{r}'(t) = \langle \cos t, \frac{1}{t}, 2 \rangle$.

Problem 25. Find $\vec{r}'(t)$ where $\vec{r}(t) = \sin(t^5)\hat{i} + \sqrt[3]{t}\hat{j} - te^t\hat{k}$